The Client: Grand Marina is a large resort facility that relies on hydrologic systems to manage water flow across multiple areas, including the main hotel building, pool and spa facilities, and kitchen/ laundry operations. The client requires a system to monitor water conditions and respond quickly to issues to ensure safety, efficiency and uninterrupted service.

The Situation: The Grand Marina needs a real- time monitoring and control system for its hydrologic infrastructure. Sensors need to continuously publish data such as pressure, flow rate every five seconds, while alerts are triggered when anomalies or leaks get detected. Operators must also be able to send commands, such as emergency shutoffs or gate adjustments, when necessary. To continue, devices must report their status regularly to ensure system reliability and connectivity. A structured messaging system is required to efficiently organize, transit and manage all data.

Topic Hierarchy: (made on google slide and then ss to put here!)

```
hydroficient/
└── grandmarina/
    ├── sensors/
    │   ├── main/
    │   │   ├── pressure_upstream
    │   │   ├── pressure_downstream
    │   │   └── flow_rate
    │   ├── pool/
    │   │   ├── pressure_upstream
    │   │   ├── pressure_downstream
    │   │   └── flow_rate
    │   └── kitchen/
    │       ├── pressure_upstream
    │       ├── pressure_downstream
    │       └── flow_rate
    │
    ├── commands/
    │   ├── main/
    │   │   ├── emergency_shutoff
    │   │   └── gate_adjustment
    │   ├── pool/
    │   │   ├── emergency_shutoff
    │   │   └── gate_adjustment
    │   └── kitchen/
    │       ├── emergency_shutoff
    │       └── gate_adjustment
    │
    ├── alerts/
    │   ├── main/
    │   │   ├── leak_detected
    │   │   ├── pressure_anomaly
    │   │   └── flow_anomaly
    │   ├── pool/
    │   │   ├── leak_detected
    │   │   ├── pressure_anomaly
    │   │   └── flow_anomaly
    │   └── kitchen/
    │       ├── leak_detected
    │       ├── pressure_anomaly
    │       └── flow_anomaly
    │
    └── status/
        ├── main/
        │   ├── heartbeat
        │   └── connection
        ├── pool/
        │   ├── heartbeat
        │   └── connection
        └── kitchen/
            ├── heartbeat
            └── connection
```

Security Reflection:

What happens if someone subscribes to hydroficient/grandmarina/#?

If someone subscribes to hydroficient/grandmarina/#, they pretty much have access to everything including sensors, alerts, commands and status across every building. Which means that they can see real time pressure and flow data every 5 seconds, any leaks or anomalies in real time, when operators send commands, and if a device is offline or not. This level of access should not be for random users since it allows too much visibility into how the system functions.

What sensitive information could they see?

They are able to see all sensitive operational data, from operational patterns to the buildings behavior to the systems structure. For example, any operational patterns they could see are, high flow in the pool which implies the pool is busy, and low activity which implies there is down time. The building behavior over time can show daily routines, and they'd be able to predict when certain areas are most or least active. The structure of the system can show topic names of how everything is organized, and if any device IDs are included, they would be able to get the exact physical locations.

What actions could they potentially take if they could also publish to command topics?

If someone can publish to command topics then there are no authentication protocols in place to get in, which is a really big issue. They would be able to trigger an emergency shutoff or play with gate adjustments. This can have serious impacts if they shut the water off, disrupt the pool or spa water supply, shut down the water to the kitchen or laundry and on top of it all if any of these happen the guests' experience may not be good.